

A Fixed-Operator Counterfactual for Overlapping Prime-Shell Labels

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Abstract

The preceding finite crosswalk found that the $Q = 60 \rightarrow 70$ transition is the unique low-correlation fracture of source-first sign labels and carries every same-prefix native-budget descent, but that crosswalk changes the physical shell/operator and the target simultaneously. We introduce a finite counterfactual that holds each full physical operator fixed and swaps only the neighboring, globally aligned label on the shell overlap; native labels are retained off the overlap. The native and transported targets are then passed through the same constrained profile-budget program at a common feasible prefix. On the locked $(N, H, z) = (512, 58, 5)$ spine, with two kernel exponents, three tolerances, and three source normalizers, this gives 18 cases and 36 operator tables. At $Q = 60 \rightarrow 70$, the right-neighbor label is cheaper on both fixed operators in 5 of 6 cases, while one case favors each home operator; all three inherited same-prefix cases are in the former class. This is a numerically certified finite target-swap atlas and a partial counterfactual control. The cross-operator interaction term, uniform asymptotics, and any twin-prime conclusion remain open.

1 Position of the question

We work on the finite source-first prime-shell line developed in the local releases TPC-302 and TPC-303 [3, 1]. TPC-302 constructs a signed label by exhaustive minimization of the physical Gram quadratic form, while TPC-303 shows that the resulting constrained native budget is not a function of shell cardinality alone on the declared spine. TPC-304 aligns labels on adjacent shell overlaps and localizes the strongest label fracture at $Q = 60 \rightarrow 70$ [2]. Its two observables, however, move together: changing Q changes both the physical operator and the source-first target.

The present question is deliberately narrower:

Within one physical operator row, what happens if the target is replaced by the neighboring shell's aligned label on the shared primes?

This is a target-swap experiment, not a claim that the two physical operators represent the same asymptotic object.

We freeze source scale $N = 512$, height $H = 58$, literal source cutoff $z = 5$, the moving shells $Q \in \{50, 60, 70, 90\}$, exponents $e \in \{1, 2\}$, tolerances $\tau \in \{1/4, 1/2, 3/4\}$, and the 17 nested prime-profile cutoffs

3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61.

All parent labels and certificates are hash-locked in the accompanying code.

2 Counterfactual protocol

Let S_L, S_R be adjacent prime shells and let $a_L : S_L \rightarrow \{-1, 1\}$ and $a_R : S_R \rightarrow \{-1, 1\}$ be their locked labels. Put $O = S_L \cap S_R$ and define

$$u = \sum_{p \in O} a_L(p) a_R(p), \quad \sigma = \begin{cases} 1, & u \geq 0, \\ -1, & u < 0. \end{cases}$$

The deterministic $+1$ choice at a zero tie fixes a convention without affecting any invariant correlation. Define full-shell transported targets

$$t_L(p) = \begin{cases} \sigma a_R(p), & p \in O, \\ a_L(p), & p \in S_L \setminus O, \end{cases} \quad t_R(p) = \begin{cases} \sigma a_L(p), & p \in O, \\ a_R(p), & p \in S_R \setminus O. \end{cases} \quad (1)$$

The left operator is consequently compared on (a_L, t_L) and the right operator on (a_R, t_R) . Equation (1) is the key scope restriction: it changes the target vector inside an operator row and does not rebuild that row.

For completeness, let V_Q denote the finite physical-output/profile matrix whose rows are indexed by S_Q , and let M be the source profile Gram matrix. For a prefix of k profiles write $V_{Q,k}$ and M_k for the corresponding submatrices. The native profile budget is

$$B_{Q,k,\tau}(b) = \min_c c^\top M_k c \quad \text{subject to} \quad \|V_{Q,k}c - b\|_2 \leq \tau \|b\|_2. \quad (2)$$

For each of $b = a_Q, t_Q$, the first least-squares-feasible prefix is recorded as $k_{Q,\text{native}}$ or $k_{Q,\text{transport}}$. Both targets are compared at

$$k = \max\{k_{Q,\text{native}}, k_{Q,\text{transport}}\}. \quad (3)$$

The TPC-302 ridge/KKT frontier evaluates (2) at this common prefix. We report the ratio

$$R_Q = \frac{B_{Q,k,\tau}(t_Q)}{B_{Q,k,\tau}(a_Q)}, \quad (4)$$

using each of three positive normalizers: $\|\beta\|_2^2$, the prefix trace mean of M , and the first profile norm. The same normalizer is used in the numerator and denominator of a row.

For a shell pair, R_L uses the left operator and R_R the right operator. We call the four strict orientation patterns

$$\begin{array}{l|l} (R_L < 1, R_R > 1) & \text{right label cheaper on both operators} \\ (R_L > 1, R_R < 1) & \text{left label cheaper on both operators} \\ (R_L > 1, R_R > 1) & \text{home operator favored} \\ (R_L < 1, R_R < 1) & \text{cross target favored.} \end{array} \quad (5)$$

An interval touching one would be unresolved rather than assigned a class.

3 Finite exact statements

Proposition 1 (Alignment and extension). *For every nonempty overlap, (1) is a well-defined binary full-shell target and maximizes the overlap inner product over the two global sign choices. A simultaneous global sign change of either parent label changes σ so that the transported vectors and all overlap mismatch counts are unchanged up to the same gauge convention.*

Proof. The two possible overlap inner products are u and $-u$, so the displayed choice of σ selects the nonnegative one, with a deterministic choice at $u = 0$. Every coordinate in (1) is a product of signs or a copied sign, hence is binary. Flipping one parent label flips u and the corresponding aligned neighboring label, leaving the aligned products invariant. The off-overlap coordinates are native by definition. $\square$

Table 1: Target-swap orientation census.

transition	left	right	home	cross	unresolved
50 $\rightarrow$ 60	4	0	0	2	0
60 $\rightarrow$ 70	0	5	1	0	0
70 $\rightarrow$ 90	3	0	2	1	0

Here “left” and “right” mean the corresponding neighboring label is cheaper on both fixed operators; “home” and “cross” are the last two strict patterns in (5).

Table 2: Central $Q = 60 \rightarrow 70$ fixed-operator ratios.

exponent	τ	R_L	R_R	orientation
1	.25	1.34649	2.02276	home
1	.50	.709463	1.07892	right
1	.75	.149193	1.90434	right
2	.25	.100203	1.15964	right
2	.50	.058495	4.78239	right
2	.75	.006411	3.84215	right

Proposition 2 (Common-prefix target comparison). *For fixed V_Q , M , and τ , the two quantities in (4) are values of the same finite constrained quadratic program with only its target vector changed. The prefix in (3) is feasible for both targets whenever their individual first feasible prefixes exist.*

Proof. Adding a profile column enlarges the coefficient space, so the least-squares residual cannot increase with k . Thus a target feasible at its first feasible prefix is feasible at every larger prefix, in particular at (3). The matrices V_Q and M are held fixed within each numerator/denominator pair; only b is replaced. $\square$

Proposition 3 (Orientation logic). *Under strict interval separation from one, the four labels in (5) are exactly the four possible combinations of which neighboring target is cheaper on the left and right fixed operators.*

Proof. On the left row, $R_L < 1$ means the aligned right target has lower budget than the native left target. On the right row, $R_R > 1$ means the aligned left target has higher budget than the native right target, so the right target is cheaper there. The other three cases follow by reversing one or both inequalities. $\square$

4 Certified finite atlas

The high-precision producer uses 55 decimal digits, 130 frontier bisection steps, and relative output enclosures of 10^{-11} . The certificate contains all labels, overlap primes, alignment signs, threshold prefixes, budgets, ratios, and classifications. The independent replay does not import the TPC-305 producer: it reconstructs the finite target protocol and checks every stored interval and orientation. The stress suite includes exhaustive small binary labels, global-sign flips, a negative-alignment fixture, and the orientation truth table.

There are 18 cases, two fixed operators per case, and three normalizers. The pair-level census is shown in Table 1.

The central fracture is the most relevant group. Table 2 reports the $\|\beta\|_2^2$ -normalized ratio intervals by their central values; the certificate stores the full 38-digit endpoint strings. The profile-trace and first-profile normalizers give the same strict side of one in every row.

The central right-label-cheaper cases include all three parent TPC-303 same-prefix descents: $(e, \tau) = (1, .50), (1, .75), (2, .75)$. This is a useful within-row target signal because those parent descents cannot be explained by an increase in the common profile dimension. It is still not a

causal proof: the left and right physical matrices differ, and the orientation changes on the outer transitions.

For scale, the six central common-prefix pairs are $(13, 15), (11, 15), (8, 9)$ for $e = 1$ and $(12, 14), (9, 10), (6, 6)$ for $e = 2$. Each pair lists the left and right maximum prefix in the certificate. These finite index changes are part of the declared protocol, not an asymptotic dimension law.

5 Interpretation and obstruction

The strongest positive result is a clean target-swap signal at the previously localized fracture: in five of six central cases, the aligned right label is cheaper on both fixed physical operators. The exact protocol makes this statement stronger than the TPC-304 coincidence because each comparison holds one operator row fixed and changes only its target.

The strongest obstruction is equally clear. A two-row table has two sources of variation: target identity and operator identity. TPC-305 controls the target variation within each row but does not make $V_L = V_R$. The outer-pair orientations also reverse or become mixed, so the central pattern cannot be promoted to a universal target-causality principle. The natural next object is the two-way holdout table: evaluate both target labels on both physical operators and form an explicit interaction contrast before assigning causal language.

Nothing here supplies arithmetic L^2 control, a fixed power saving, a full Route-B Gate-B theorem, or a twin-prime result. The Session-named Route-A and Route-B evaluator files are absent from this checkout; the claim level is the local fail-closed level recorded in the route-evaluation note.

Reproducibility

The canonical certificate is `results/tpc305_certificate.json`. The producer, independent checker, stress suite, and Bridge-B checker are all included in the project directory. Parent TPC-302 and TPC-303 source/certificate hashes are embedded in the certificate. The manuscript is intentionally a finite research report: no claim is made that the observed orientations persist as Q or the source scale grows.

References

- [1] Liang Wang. A cardinality-monotonicity obstruction for finite native budgets, 2026. Local TPC-303 release in the prime-dynamics research repository.
- [2] Liang Wang. Overlapping prime-shell sign transport localizes a finite native-budget descent, 2026. Local TPC-304 release in the prime-dynamics research repository.
- [3] Liang Wang. Source-first sign enumeration and a growing-shell budget-gap audit, 2026. Local TPC-302 release in the prime-dynamics research repository.